Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Equation of chord (AB) of circle $x^2+y^2=2$ passing through the point $P(2,2)$, such that $\frac{PB}{PA}=3$

(A) $x=3y$

(B) $x=y$

(C) $y-2=\sqrt{3}(x-2)$

(D) none of these

Q.2

The image of the centre of the circle $x^2+y^2=2a^2$ with respect to the line $x+y=1$ is

(A) $(\sqrt{2},\sqrt{2})$

(B) $\left(\frac{1}{\sqrt{2}},\sqrt{2}\right)$

(C) $\left(\sqrt{2},\frac{1}{\sqrt{2}}\right)$

(D) none of these

Q.3

The condition that the line $lx+my+n=0$ be normal to the circle $x^2+y^2+2gx+2fy+c=0$ is

(A) $lm=gf$

(B) $lg+nf=m$

(C) $lg+mf=n$

(D) none of these

Q.4

The area of the circle centred at $(1,2)$ and passing through the point $(4,6)$ is

(A) 5π

(B) 10π

(C) 25π

(D) none of these

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Q.5

If the chord of contact of the tangents from a point on the circle $x^2 + y^2 = a^2$ to the circle $x^2 + y^2 = b^2$ touch the circle $x^2 + y^2 = c^2$, then the roots of the equation $ax^2 + 2bx + c = 0$ are necessarily

(A) imaginary

(B) real and equal

(C) real and unequal

(D) rational

Q.6

The triangle PQR is inscribed in the circle $x^2 + y^2 = 25$. If Q and R have coordinates $(3, 4)$ and $(-4, 3)$ respectively, then $\angle QPR$ is equal to

(A) $\pi/2$

(B) $\pi/3$

(C) $\pi/4$

(D) $\pi/6$

Q.7

If the circles $x^2 + y^2 + 2x + 2ky + 6 = 0$ and $x^2 + y^2 + 2ky + k = 0$ intersect orthogonally, the k is

(A) 2 or $-3/2$

(B) -2 or $-3/2$

(C) 2 or $3/2$

(D) -2 or $3/2$

Q.8

The equation of a circle whose centre lies on $3x - y - 4 = 0$ and $x + 3y + 2 = 0$ and has an area 154 square units is

(A) $x^2 + y^2 - 2x + 2y - 47 = 0$

(B) $x^2 + y^2 + 2x - 2y - 47 = 0$

(C) $x^2 + y^2 - 2x + 2y + 47 = 0$

(D) none of these

Q.9

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If two lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ cut the coordinate axes at concyclic points, then

(A) $a_1a_2 + b_1b_2 = 0$

(B) $a_1a_2 - b_1b_2 = 0$

(C) $a_1b_1 + a_2b_2 = 0$

(D) $a_1b_1 - a_2b_2 = 0$

Q.10

The locus of the point of intersection of the tangents to the circle $x = r\cos\theta, y = r\sin\theta$ at points whose parametric angles are differ by $\pi/3$, is

(A) $x^2 + y^2 = 4(2 - \sqrt{3})r^2$

(B) $3\left(x^2 + y^2\right) = 1$

(C) $x^2 + y^2 = (2 - \sqrt{3})r^2$

(D) $3\left(x^2 + y^2\right) = 4r^2$

Q.11

The point $(1, 4)$ lies inside the circle $x^2 + y^2 - 6x - 10y + p = 0$ which does not touch or intersect the coordinate axes, then

(A) $0 < p < 29$

(B) $25 < p < 29$

(C) $9 < p < 25$

(D) $9 < p < 29$

Q.12

The equation of the circle which passes through $(2a, 0)$ and has the radical axis $2x - a = 0$ with the circle $x^2 + y^2 = a^2$ is

(A) $x^2 + y^2 - 2ax = 0$

(B) $x^2 + y^2 + 2ax = 0$

(C) $x^2 + y^2 + 3ax = 0$

(D) $x^2 + y^2 - 3ax = 0$

Q.13

A variable circle always touches the line $y = x$ and passes through the point $(0, 0)$. The common chords of above circle and $x^2 + y^2 + 6x + 8y - 7 = 0$ will pass through a fixed point, whose coordinates are

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(A) $\left(\frac{1}{2}, \frac{1}{2}\right)$

(B) $\left(\frac{1}{2}, -\frac{1}{2}\right)$

(C) $\left(-\frac{1}{2}, -\frac{1}{2}\right)$

(D) $\left(-\frac{1}{2}, \frac{1}{2}\right)$

Q.14

Two rods of lengths a and b slide along the x-axis and y-axis respectively in such a manner that their ends are concyclic. The locus of the centre of the circle passing through the end points is

(A) $4\left(x^2+y^2\right)=a^2+b^2$

(B) $x^2+y^2=a^2+b^2$

(C) $4\left(x^2-y^2\right)=a^2-b^2$

(D) $x^2-y^2=a^2-b^2$

Q.15

The equation of the circle, which is touched by the line $y=x$, has its centre on the positive direction of the x-axis and cut off a chord of length 2 units on the line $\sqrt{3}y=x$, is

(A) $x^2+y^2-4x+2=0$

(B) $x^2+y^2-4x+1=0$

(C) $x^2+y^2-8x+8=0$

(D) $x^2+y^2-4y+2=0$

Q.16

Consider a family of circles which are passing through the point $(-1,1)$ and are tangent to x-axis. If (h,k) are the co-ordinates of the centre of the circles, then the set of values of k is given by the interval

(A) $0<k<\frac{1}{2}$

(B) $k \geq \frac{1}{2}$

(C) $-\frac{1}{2} \leq k \leq \frac{1}{2}$

(D) $k \leq \frac{1}{2}$

Q.17

The locus of the centre of a circle of radius 2 which rolls outside on the circumference of circle $x^2+y^2+3x-6y-9=0$ is

(A) $x^2+y^2+3x-6y+5=0$

(B) $x^2+y^2+3x-6y-31=0$

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(C) $x^2+y^2+3x-6y+\frac{19}{4}=0$

(D) none of these

Q.18

An equation of the chord of the circle $x^2+y^2=a^2$ passing through the point $(2,3)$ farthest from the centre is

(A) $2x+3y=13$

(B) $3x-y=3$

(C) $x-2y+4=0$

(D) $x-y+1=0$

Q.19

If the circumference of the circle $x^2+y^2+8x+8y-b=0$ is bisected by the circle $x^2+y^2-2x+4y+a=0$, then $a+b=$

(A) 50

(B) 56

(C) -56

(D) -34

Q.20

Two fixed circles with radii r_1 and r_2 $(r_1>r_2)$ respectively touch each other externally. Then the locus of the point of intersection of their direct common tangents is

(A) a circle

(B) a straight line

(C) a parabola

(D) none of these

Q.21

The square of the area of the triangle formed by the positive x-axis and the normal and tangent to the circle $x^2+y^2=4$ at $(1,\sqrt{3})$ is

Q.22

If the curves $ax^2+4xy+2y^2+x+y+5=0$ and $ax^2+6xy+5y^2+2x+3y+8=0$ intersect at four concyclic points then the value of $|a|$ is

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Q.23

A line is drawn through the point $P(3, 11)$ to cut the circle $x^2 + y^2 = 9$ at A and B. Then $PA.PB$ is equal to

Q.24

Let $2x^2 + y^2 - 3xy = 0$ be the equation of a pair of tangents drawn from the origin O to a circle of radius 3 with centre in the first quadrant. If A is one of the points of contact. If $OA = \lambda + \sqrt{\mu}$, then the find the value of $\lambda^2 + \mu$

Q.25

The equation $\lambda^2 x^2 + \left(\lambda^2 - 5\lambda + 4\right) xy + (3\lambda - 2)y^2 - 8x + 12y - 4 = 0$ will represent a circle, if λ is equal to

Q.26

The angle in degrees between the tangents drawn from the origin to the circle $(x - 7)^2 + (y + 1)^2 = 25$ is

Q.27

If the straight line $y = mx$ is outside the circle $x^2 + y^2 - 20y + 90 = 0$, then the largest positive integral value of m is

Q.28

The area of the quadrilateral formed by the tangents from the point $(3, 4)$ to the circle $x^2 + y^2 = 9$ and the line joining their point of contact to the centre, is

Q.29

The length of the intercept on the straight line $4x - 3y - 10 = 0$ by the circle $x^2 + y^2 - 2x + 4y - 20 = 0$ is

Q.30

If a circle passes through the point $(1, 2)$ and cuts the circle $x^2 + y^2 = 4$ orthogonally, then the equation of the locus of its centre is $2x + 4y = m$ where $m =$

Answers & Solutions

Q.1 Answer:

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$x = y$

Solution:

As $\frac{PB}{PA} = 3 \quad \Rightarrow \quad AB = (PB - PA) = 2\sqrt{2} \quad \{Q \quad (PA)(PB) = PT^2\}$

As point P is $(2, 2)$ and the given circle $x^2 + y^2 = 2$ has diameter $2\sqrt{2}$.

$\Rightarrow \quad AB$ is diameter of the circle.

$\Rightarrow$ Line must pass through the centre $(0, 0)$ also.

Q.2 Answer:

none of these

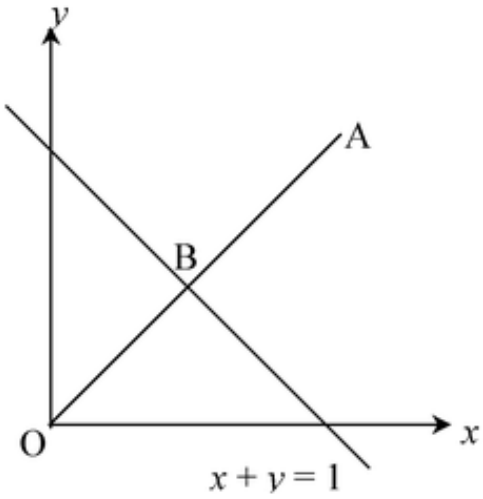
Solution:

Centre of the circle is $(0, 0)$

Let A is the image of the origin in the line $x + y = 1$

$$OB = \frac{1}{\sqrt{2}} \Rightarrow OA = \sqrt{2}$$

$$\therefore \quad A \equiv (1, 1)$$



Q.3 Answer:

$lg + mf = n$

Solution:

Every normal of circle passes through centre of circle.

$\therefore$ If $\mathrm{lx + my + n = 0}$ is normal to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$, then $(-g, -f)$ lie on $\mathrm{lx\ my + n = 0}$

so, $\mathrm{1(-g) + m(-f) + n = 0}$

$\mathrm{lg + mf - n = 0}$

Q.4 Answer:

25π

Solution:

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The equation of a circle centred at $(1,2)$ and passing through $(4,6)$ is

$$(x-1)^2+(y-2)^2=(4-1)^2+(6-2)^2$$

$$\Rightarrow x^2+y^2-2x-4y+1+4=9+16$$

$$\Rightarrow x^2+y^2-2x-4y-20=0$$

Now, radius $=\sqrt{(-1)^2+(-2)^2+20}$

$=\sqrt{1+4+20}==\sqrt{1+4+20}=5$

$\therefore$ **Area of circle** $=\pi r^2$

$=25\pi$ **sq unit**

Q.5 Answer:

real and equal

Solution:

Let $P(a\cos\theta, b\sin\theta)$ **be a point on** $x^2+y^2=a^2$

Chord of contact of tangents from P **to circle** $x^2+y^2=b^2$ **is**

$$xa\cos\theta+ya\sin\theta=b^2$$

$$\Rightarrow y=-\cot\theta x+\frac{b^2}{a}\operatorname{cosec}\theta \quad ...(i)$$

Since line (i) is tangent to circle $x^2+y^2=c^2$

$$\therefore \quad \left(\frac{b^2}{a}\operatorname{cosec}\theta\right)^2=c^2\left(1+\cot^2\theta\right)\Rightarrow b^2=ac$$

But $D=4b^2-4ac=4ac-4ac=0$

$\therefore$ **Roots of equation** $ax^2+2bx+c=0$ **are real and equal**

Q.6 Answer:

$\pi/4$

Solution:

Now, length of chord $QR=$

$\sqrt{(-4-3)^2+(3-4)^2}=5\sqrt{2}$

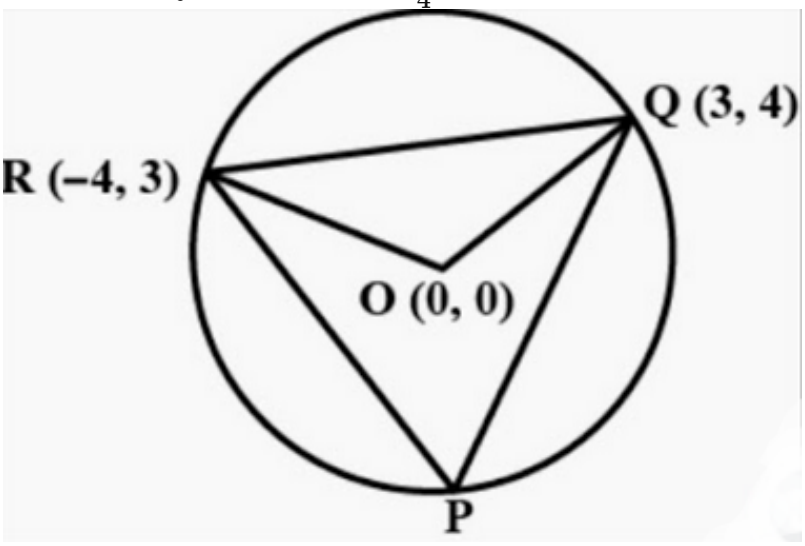
Also, $OR=OQ=5$ **(radii of the circle)**

$\therefore \angle ROQ=90^\circ$

Using the theorem that arc subtended by a chord is twice the angle subtended by the chord at the arc of the circle,

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we have $\angle QPR = 45^\circ = \frac{\pi}{4}$



Q.7 Answer:

2 or $-3/2$

Solution:

$2.(1)(0) + 2(k) = k + 6$
or $2k^2 - k - 6 = 0$
or $2k^2 - 4k + 3k - 6 = 0$
or $k = -3/2, +2$

Q.8 Answer:

$x^2 + y^2 - 2x + 2y - 47 = 0$

Solution:

The centre of the circle is the point of intersection of the given lines i.e. the point $(1, -1)$**. If** r **is the radius of the circle, then its area** $\pi r^2 = 154$
$\therefore$ **Equation of the circle is** $(x-1)^2 + (y+1)^2 = 7^2$
$\Rightarrow \quad x^2 + y^2 - 2x + 2y = 47$

Q.9 Answer:

$a_1a_2 - b_1b_2 = 0$

Solution:

The given lines meet x**-axis at** $(-c_1/a_1, 0)$ **and** $(-c_2/a_2, 0)$ **,** y**-axis at** $(0, -c_1/b_1)$ **and** $(0, -c_2/b_2)$
If the equation of the circle passing through these points is $x^2 + y^2 + 2gx + 2fy + c = 0$**, then** $-c_1/a_1, -c_2/a_2$ **are the roots of** $x^2 + 2gx + c = 0 \Rightarrow c = c_1c_2/a_1a_2$
Similarly $c = c_1c_2/b_1b_2 \Rightarrow a_1a_2 - b_1b_2 = 0$

Q.10 Answer:

$3\left(x^2 + y^2\right) = 4r^2$

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Solution:

Let one of the points on the circle be $P(r\cos\theta, r\sin\theta)$.

Then the other point is $Q\{r\cos(\theta+\pi/3), r\sin(\theta+\pi/3)\}$.

$\therefore$ Equation of the tangent at P is $x\cos\theta + y\sin\theta = r, \ldots$ (i)

and the equation of the tangent at Q is $x\cos\left(\theta+\frac{\pi}{3}\right) + y\sin\left(\theta+\frac{\pi}{3}\right) = r$

$\Rightarrow \left[\cos\theta\left(\frac{1}{2}\right) - \sin\theta\left(\frac{\sqrt{3}}{2}\right)\right] + y\left[\sin\theta\left(\frac{1}{2}\right) + \cos\theta\left(\frac{\sqrt{3}}{2}\right)\right] = r$

$\Rightarrow \frac{1}{2}(x\cos\theta + y\sin\theta) - \frac{\sqrt{3}}{2}(x\sin\theta - y\cos\theta) = r$

$\Rightarrow \quad \frac{r}{2} - \frac{\sqrt{3}}{2}(x\sin\theta - y\cos\theta) = r \quad (x\sin\theta - y\cos\theta) = -\frac{r}{\sqrt{3}}$

The locus of the point of intersection of these tangent can now be obtained by eliminating θ

$\therefore$ Squaring and adding (i) and (ii)

$\Rightarrow \quad (x\cos\theta + y\sin\theta)^2 + (x\sin\theta - y\cos\theta)^2 = r^2 + \left(-\frac{r}{\sqrt{3}}\right)^2$

$\Rightarrow \quad 3\left(x^2+y^2\right) = 4r^2$

Q.11 Answer:

$25 < p < 29$

Solution:

Since the circle does not touch or intersect the coordinates axes, the absolute values of x and y coordinates of the centre are greater than the radius of the circle. Coordinates of the centre of the circle are $(3, 5)$ and the radius is $\sqrt{9+25-p}$

so that

$3 > \sqrt{9+25-p} \quad \Rightarrow p > 25$ ...(i)

$5 > \sqrt{9+25-p} \quad \Rightarrow p > 9$ ....(ii)

$\therefore$ Point $(1, 4)$ lies inside the circle $x^2 + y^2 - 6x - 10y + p = 0$

$\Rightarrow 1 + 16 - 6 - 10 \times 4 + p < 0 \Rightarrow p < 29$ ...(iii)

From (i), (ii), (iii) we get $25 < p < 29$.

Q.12 Answer:

$x^2 + y^2 - 2ax = 0$

Solution:

Let the equation of circle be $\mathrm{x}^2 + \mathrm{y}^2 + 2\,\mathrm{gx} + 2\mathrm{fy} + \mathrm{c} = 0$

Since, it passes through the point $(2\mathrm{a}, 0)$, so $4a^2 + 4ag + c = 0$

Also, its radical axis with $x^2 + y^2 = a^2$ is $2gx + 2fy + c + a^2 = 0$

But radical axis is $\mathrm{x} = \frac{\mathrm{a}}{2}$, so we get $\frac{2\,\mathrm{g}}{1} = \frac{\mathrm{c}+\mathrm{a}}{-\mathrm{a}/2}$

or $\mathrm{ag} + \mathrm{c} + \mathrm{a} = 0$

From equations (i) and (ii), we get g and c Hence, the equation is $\mathrm{x}^2 + \mathrm{y}^2 - 2\mathrm{ax} = 0$.

Q.13 Answer:

$\left(\frac{1}{2}, \frac{1}{2}\right)$

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Solution:

Family of circles $x^2+y^2+\lambda(x-y)=0$
For common chords $S_1-S_2=0 \Rightarrow 6x+8y-7-\lambda(x-y)=0$
Point of concurrency is intersection point of lines $6x+8y-7=0$ **and** $x-y=0$
$\Rightarrow (x,y)=\left(\frac{1}{2},\frac{1}{2}\right)$

Q.14 Answer:

$4\left(x^2-y^2\right)=a^2-b^2$

Solution:

Let $C(h,k)$ **be the centre of the circle passing through the end points of the rod** AB **and** PQ **of lengths** a **and** b **respectively,** CL **and** CM **be perpendiculars from** C **on** AB **and** PQ **respectively.**

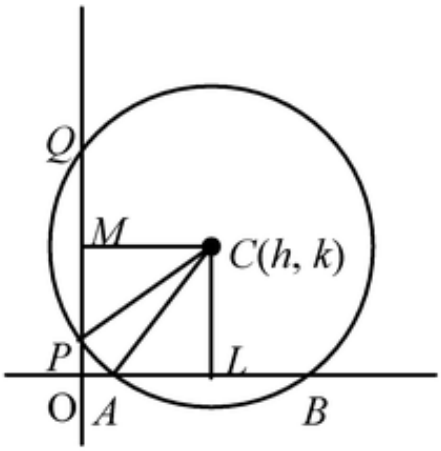
$\therefore \quad AL=\frac{a}{2} \quad BM=\frac{b}{2}$

then $CA=CP$ **(radii of the same circle)**

$\Rightarrow \quad k^2+\frac{a^2}{4}=h^2+\frac{b^2}{4}$

$\Rightarrow \quad 4\left(h^2-k^2\right)=a^2-b^2$

so that locus of (h,k) **is** $4\left(x^2-y^2\right)=a^2-b^2$



Q.15 Answer:

$x^2+y^2-4x+2=0$

Solution:

$OP=CP=\frac{h}{\sqrt{2}}=AC$

$CM=h\sin 30^\circ=\frac{h}{2}$

From ΔACM

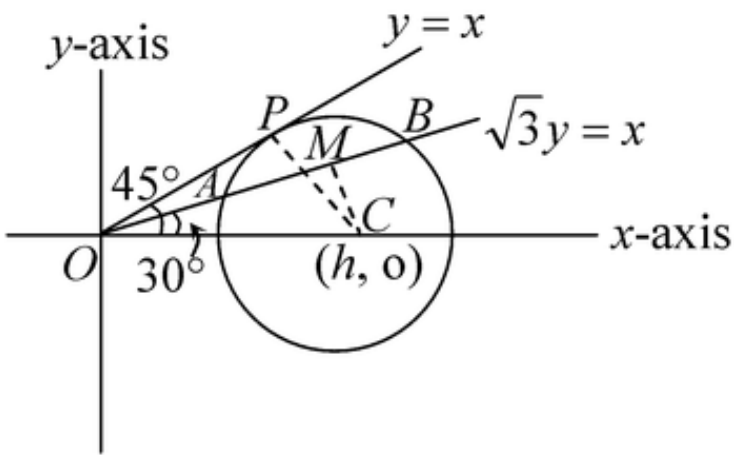
$AC^2=AM^2+CM^2$

$\frac{h^2}{2}=1+\frac{h^2}{4} \Rightarrow \quad \frac{h^2}{4}=1 \Rightarrow \quad h=2$

radius $=\sqrt{2}$

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$(x-2)^2+y^2=2; x^2+y^2-4x+2=0$



Q.16 Answer:

$k \geq \frac{1}{2}$

Solution:

Centre of circle is (h,k) and x-axis is tangent. $\Rightarrow$ Radius of the family of circle $= |k|$
Also circle passes through $(-1,1) \Rightarrow (1+h)^2+(1-k)^2=k^2$
$\Rightarrow \quad h^2+2h+2-2k=0$ for real $h, \quad k \geq \frac{1}{2}$

Q.17 Answer:

$x^2+y^2+3x-6y-31=0$

Solution:

If centre of the circle which rolls outside the given circle be (h,k), then distance between the centres $=$ sum of radii.
$\Rightarrow (h+3/2)^2+(k-3)^2=\left(\frac{9}{2}+2\right)^2 \Rightarrow h^2+k^2+3h-6k-31=0$

Q.18 Answer:

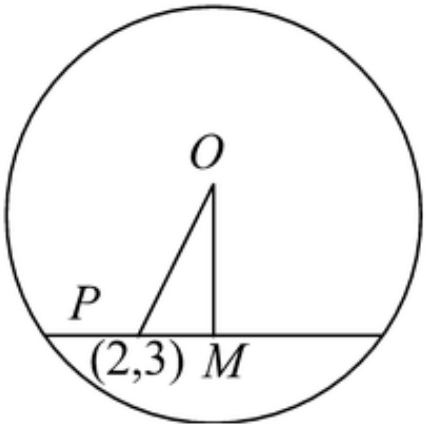
$2x+3y=13$

Solution:

Let $P(2,3)$ be the given point, M be the middle point of a chord of the circle $x^2+y^2=a^2$ through P. Then the distance of the centre O of the circle from the chord is OM
and $(OM)^2=(OP)^2-(PM)^2$
which is maximum when PM is minimum, i.e. M coincides with P i.e. P is the middle point of the

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chord.



Hence the equation of the chord $(T = S_1)$ **is**

$2 \cdot x + 3 \cdot y - a^2 = (2)^2 + (3)^2 - a^2 \Rightarrow 2x + 3y = 13$

Q.19 Answer:

-56

Solution:

Equation of radical axis (i.e. common chord) of the two circles is $S_1 - S_2 = 0$

$\Rightarrow \quad 10x + 4y - a - b = 0$ **...(i)**

Centre of first circle is $C_1(-4, -4)$

Since second circle bisects the circumference of the first circle, therefore, centre $C_1(-4, -4)$ **of the first circle must lie on the common chord (i)**

$\therefore -40 - 16 - a - b = 0 \Rightarrow a + b = -56$

Q.20 Answer:

a straight line

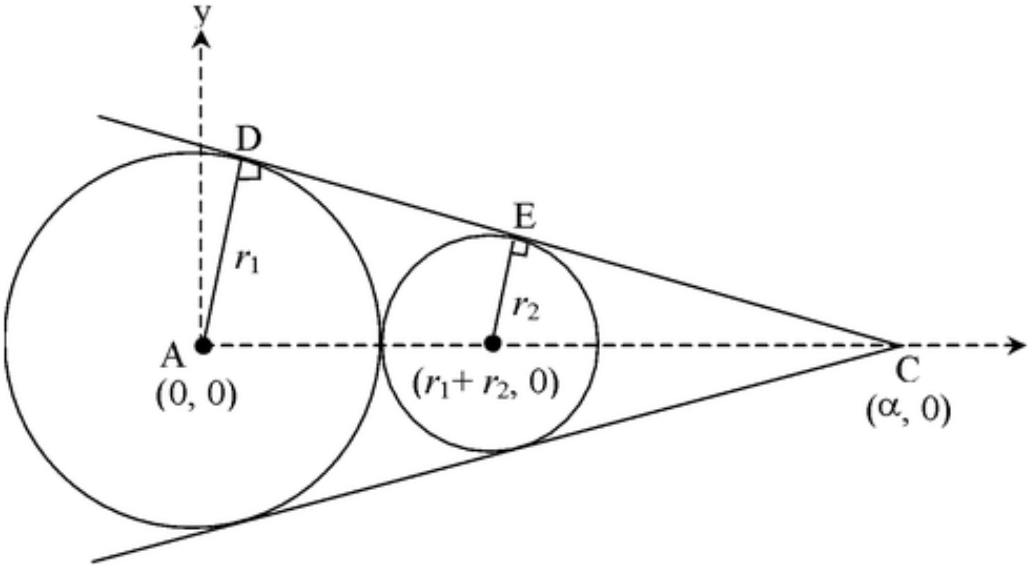
Solution:

Let $A \equiv (0, 0)$ **and** $B \equiv (r_1 + r_2, 0)$ **be the centres of the two given fixed circles. Let** $C \equiv (\alpha, 0)$ **be the point of intersection of direct common tangents.**

Now, $\frac{r_2}{r_1} = \frac{\alpha - (r_1 + r_2)}{\alpha} \Rightarrow r_2\alpha = r_1\alpha - r_1^2 - r_1r_2 \Rightarrow \alpha = \frac{r_1^2 + r_1r_2}{r_1 - r_2}$

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Locus of C is $x = \frac{r_1^2 + r_1 r_2}{r_1 - r_2} = a$ constant, which is a straight line.



Q.21 Answer:

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Solution:

Equation of tangent at the point $P(1, \sqrt{3}) \equiv x + y\sqrt{3} = 4$
Equation of normal at the point $P(1, \sqrt{3}) \equiv x\sqrt{3} - y = 0$
These lines cut the x axis at $A(4, 0)$ and $B(0, 0)$
Hence area of ΔPAB is $\frac{1}{2}\begin{vmatrix} 1 & 1 & \sqrt{3} \\ 1 & 4 & 0 \\ 1 & 0 & 0 \end{vmatrix} = 2\sqrt{3}$

Q.22 Answer:

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Solution:

Any second degree curve passing through the intersection of the given curves is
$ax^2 + 4xy + 2y^2 + x + y + 5 + \lambda\left(ax^2 + 6xy + 5y^2 + 2x + 3y + 8\right) = 0$
If it is a circle, then coefficient of x^2 = coefficient of y^2 and coefficient of $xy = 0$
$a(1 + \lambda) = 2 + 5\lambda$ and $4 + 6\lambda = 0$
$\Rightarrow \quad a = \frac{2 + 5\lambda}{1 + \lambda} \quad$ and $\quad \lambda = -\frac{2}{3}$
$\Rightarrow \quad a = \frac{2 - \frac{10}{3}}{1 - \frac{2}{3}} = -4$

Q.23 Answer:

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Solution:

From geometry we know $PA.PB = (PT)^2$, where PT is the length of the tangent from P to the circle.

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Hence $PA.PB = (3)^2 + (11)^2 - 9 = 11^2 = 121$

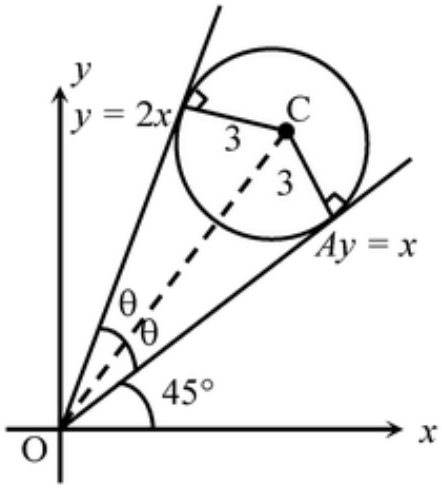
Q.24 Answer:

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Solution:

$2x^2 + y^2 - 3xy = 0$

$\Rightarrow \quad (2x - y)(x - y) = 0 \Rightarrow y = 2x, y = x$



are the equation of straight lines passing through origin.

Now let the angle between tangents is 2α**, then** $\tan(45^\circ + 2\alpha) = 2$

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$$\Rightarrow \frac{\tan 45^\circ + \tan 2\alpha}{1 - \tan 45^\circ \tan 2\alpha} = 2 \quad \Rightarrow \quad \frac{1 + \tan 2\alpha}{1 - \tan 2\alpha} = \frac{2}{1}$$

$$\Rightarrow \frac{2\tan 2\alpha}{2} = \frac{1}{3}$$

(By componendo and dividendo rule)

$\Rightarrow \quad \frac{2\tan\alpha}{1-\tan^2\alpha} = \frac{1}{3} \Rightarrow \tan^2\alpha + 6\tan\alpha - 1 = 0$

$\tan\alpha = \frac{-6 \pm \sqrt{(36+4)}}{2} = -3 \pm \sqrt{10} = -3 + \sqrt{10} \quad (\square 0 < \alpha < \frac{\pi}{4})$

Now, in $\Delta OAC, \tan\alpha = \frac{3}{OA} = (\sqrt{10} - 3)$

$\therefore \quad OA = \frac{3(\sqrt{10}+3)}{(\sqrt{10}-3)(\sqrt{10}+3)} = 3(3 + \sqrt{10}) = 9 + \sqrt{90} = \lambda + \sqrt{\mu}$

$\therefore \quad \lambda = 9$ **and** $\mu = 90$**, then** $\lambda^2 + \mu = 81 + 90 = 171$

Q.25 Answer:

—

Solution:

The given equation will represent a circle, if

coefficient of x^2 **= coefficient of** y^2 **and coefficient of** $xy = 0$

$\Rightarrow \lambda^2 = 3\lambda - 2$ **and** $\lambda^2 - 5\lambda + 4 = 0 \Rightarrow \lambda^2 - 3\lambda + 2 = 0$ **and** $\lambda^2 - 5\lambda + 4 = 0$

$\Rightarrow (\lambda - 1)(\lambda - 2) = 0$ **and** $(\lambda - 1)(\lambda - 4) = 0 \Rightarrow \lambda = 1$

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Q.26 Answer:

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Solution:

Suppose $y = mx$ be a tangent from the origin to the circle
$(x-7)^2 + (y+1)^2 = 5^2$, **then** $\frac{7m-(-1)}{\sqrt{m^2+1}} = \pm 5$
Let m_1 and m_2 be the slopes of the tangents. Then $m_1 m_2 = -\frac{12}{12}$
Hence the angle between two tangents is $\frac{\pi}{2}$

Q.27 Answer:

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Solution:

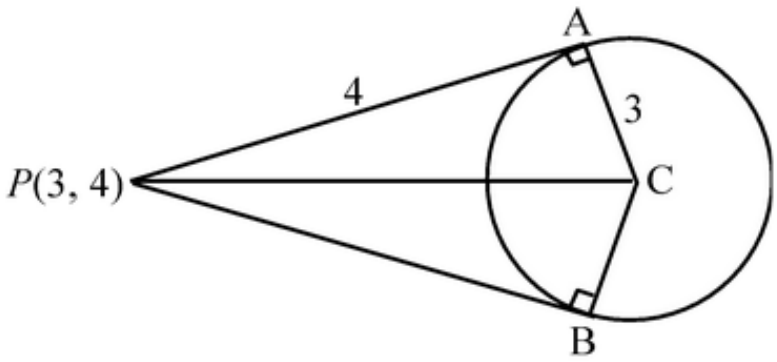
$\frac{10}{\sqrt{1+m^2}} > \sqrt{10}$ **or** $\left(1+m^2\right) < 10$ **or** $m^2 < 9$ **or** $|m| < 3$

Q.28 Answer:

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Solution:

Length of the tangents from the point $P(3,4)$ to the circle $x^2 + y^2 = 9$ is
$PA = PB = \sqrt{(3)^2 + (4)^2 - 9} = 4$
$\therefore$ Area of $\Delta PAB = 2($ area of $\Delta PAC)$
$= 2 \times \frac{1}{2} \times 3 \times 4 = 12$



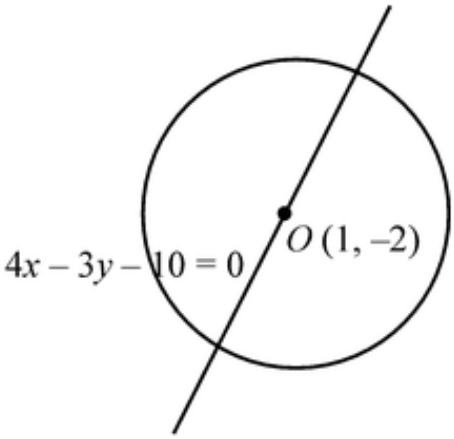
Q.29 Answer:

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Solution:

Centre and radius of circle $x^2 + y^2 - 2x + 4y - 20 = 0$ are $(1, -2)$ and $\sqrt{(1+4+20)} = 5$ respectively.

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Let OM be the perpendicular from O on the line $4x - 3y - 10 = 0$, then

$OM = \frac{|4\times1-3\times(-2)-10|}{\sqrt{4^2+(-3)^2}} = 0$

Hence line $4x - 3y - 10 = 0$ passes through the centre of the circle. Hence intercepted length = diameter of the circle

$= 2 \times 5 = 10$

Q.30 Answer:

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Solution:

Let the circle be $x^2 + y^2 + 2gx + 2fy + c = 0$. This passes through $(1, 2)$, therefore $5 + 2g + 4f + c = 0$ ...(i)

The circle $x^2 + y^2 = 4$ intersects the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ orthogonally. Therefore,

$2(g \cdot 0 + f \cdot 0) = c - 4 \Rightarrow c = 4$

Putting $c = 4$ in (i), we get $2g + 4f + 9 = 0$

Therefore, the locus of $(-g, -f)$ is $-2x - 4y + 9 = 0$

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